

The Good, the Bad and the Ugly

TEXT MINING AND SENTIMENT ANALYSIS

Master in Data Science and Economics

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Introduction

In recent years, streaming services have seen explosive growth, attracting millions of viewers who engage with content daily. These platforms, particularly during live broadcasts, allow viewers to interact directly with streamers and with each other, sharing emotions, opinions, and reactions in real time. For popular streams, a single hour can yield hundreds of thousands of messages, capturing a rich spectrum of viewer sentiment and topical engagement.

This project focuses on analyzing Twitch chat data from the 2021 Counter-Strike esports event "[PGL Major Stockholm 2021](#)", the most viewed tournament in the discipline, which took place over 12 gaming days and generated approximately 1,670,000 chat messages on the main broadcast channel - PGL. The project aims to shed light on the unique nature of real-time viewer interactions in Twitch streams and the broader context of engagement on streaming platforms. The dataset is provided by Esport Charts.

There are several most common approaches for sentiment analysis.

Lexical Approaches. For example, keyword-based approaches are easily implementable and can achieve good accuracy values. However, this approach has several drawbacks:

- Determining the content of the emotion lexicon is subjective.
- It may result in incorrect recall values.
- Selected words might be ambiguous [Suttles, 2013].
- Moreover, keyword-based approaches are not suitable for a wide range of domains.

Ontology-Based Approaches Ontology-based approaches allow the use of commonsense knowledge, improving recall values. However, creating an emotional ontology is a tedious and time-consuming task. As a result, lexical resources are often used as features in Machine Learning algorithms.

Machine Learning Approaches. Supervised learning is widely used for emotion detection because it generally yields better results compared to unsupervised learning [Kim, 2011].

However, supervised approaches require:

- Labelling of training examples.
- Annotating of examples.

I would like to describe emotions used in emotional detection systems.

Ekman's Basic Emotion Model [Ekman, 1999]. This model identifies **six basic emotions**: Anger, Disgust, Fear, Happiness, Sadness, Surprise.

Plutchik's Emotion Model [Plutchik, 1980]. Plutchik expands Ekman's model and defines **eight basic bipolar emotions**, which include: **Anger** vs. **Fear**, **Joy** vs. **Sadness**, **Trust** vs. **Disgust**, **Surprise** vs. **Anticipation**

Russell's Circumplex Model of Affect [Russell, 1980]: Emotions are represented in a two-dimensional space: **Valence Dimension**: Indicates whether an emotion is **Pleasant** or

Unpleasant. Arousal Dimension: Differentiates between **Activation** and **Deactivation** states.

Mehrabian's PAD Model [Mehrabian, 1996]. Emotions are described using a three-dimensional representation: **Pleasure (P)**: Pleasantness of an emotion. **Arousal (A)**: Level of activation. **Dominance (D)**: Whether the subject feels in control of the situation.

Both **categorical** and **dimensional** models have their benefits and limitations, and the choice of a model depends on the emotions you aim to detect.

Research question and methodology

With the rapid development of esports, many stakeholders — including brands, tournament organizers, agencies, and streamers — are increasingly seeking to better understand the space they are performing in. To achieve this, everything that can be measured should be measured. Sentiment analysis of livestreams is one of the least explored areas in streaming and esports. Yet, it holds great potential in helping stakeholders better understand viewer moods, preferences, and emotional reactions — all of which are key to achieving their goals.

These goals might include: improving brand integration, enhancing the quality of broadcasts, boosting audience engagement, and more.

The core challenge and formal definition of the problem lies in developing a deeper understanding of the audience and their real-time reactions.

Hence, the goal of this project is to analyze and evaluate the sentiment during live-streamed esports event.

To achieve this goal, I would like to try different approaches to understand and evaluate the sentiment of the event. Sentiment analysis of esports event can be based on:

- Twitch chat sentiment
- Annotator sentiment
- Viewers activity

Twitch chat sentiment

Twitch chat is a special place on the Twitch platform, where the audience/viewers of the live broadcasting can interact with each other, share their opinions and react on what is happening at live-broadcast. The peculiarity of Twitch chat is the fact that it can be visible and it is visible only for the audience who does not hide it from the screen. Important to mention that only the share of the total audience interacted through the chat. It is also called the active audience / engaged audience / active chatters.

Twitch chat provides a unique dataset due to its fast-paced, highly interactive nature, where messages often include short text, emotes, slang, and jargon. The sentiment of viewers, as expressed in these messages, is strongly influenced by events occurring on the screen, and its

analysis requires identifying trends, understanding shifts, and comparing multiple computational approaches to achieve reliable results.

The object in analysing Twitch chat sentiment is one particular message.

Twitch chat provides a unique dataset due to its fast-paced, highly interactive nature, where messages often include short text, emotes, slang, spam and jargon. The sentiment of viewers, as expressed in these messages, is strongly influenced by events occurring on the screen, and its analysis requires identifying trends, understanding shifts, and comparing multiple computational approaches to achieve reliable results. Also, it is important to mention, that twitch chat represent sentiment of the active and engaged audience, not all viewers.

Annotator sentiment

Every livestream or broadcast is typically accompanied by commentary — either by the streamer (in the case of general streaming) or by professional casters (in the case of esports).

In streaming, this commentary often takes the form of direct interaction with the audience, sharing personal opinions and reactions to what's happening on screen. The tone and content of the commentary largely depend on the category in which the streamer operates.

In IRL (In Real Life) streams, the discussions can revolve around any social or everyday topics. In gaming streams, the conversation is usually more focused on game mechanics, gameplay strategy, or reactions to in-game events.

In contrast, esports broadcasts resemble traditional sports coverage. Commentary is usually delivered by experts or pro players, who analyze gameplay in real time and post-match — breaking down mistakes, highlighting successful plays, and identifying key moments.

These broadcasts also often include brand integrations, interviews, and other segments that vary depending on the tournament organizer.

The object in analysing Twitch annotator's sentiment is the phrase that was told by annotator.

The formal problem in this is a transcription of the speech, slang and jargon. Moreover, the annotators themselves can cheer for some particular team, which somehow makes their speech biased. Additionally, the match itself can be extremely boring, while the annotators will try to make it more dynamic positive etc.

Viewers activity

This approach is based not on the speech or text, but on the statistical metrics.

For example, we can measure the number of viewers at each moment during a broadcast. By analyzing how viewership fluctuates over time, we can estimate audience interest — the more engaging the match, the more viewers stay, and the more new viewers join.

An alternative set of metrics includes:

- the number of messages per time interval,
- the number of active chatters,
- the percentage of active chatters relative to total viewers,
- the average message length.

These indicators can provide deeper insight into viewer engagement and interaction quality during a stream. Object is the metric at a particular moment of time.

Approach

Goal: Analyze and evaluate the sentiment during live-streamed esports event:

- Analyze sentiment of Twitch chat
- Analyze sentiment of annotators
- Analyze the main metrics fluctuation
- Compare the sentiment across different aspects with the real video.

Sentiment of Twitch chat

Due to the nature of Twitch, I believe that a model based on emotional dimensions will work well. There are hundreds of messages per minute during a broadcast. This means that during the aggregation of sentiment from individual messages to the sentiment of a specific minute, we need an effective method to process and interpret the data.

Imagine that I use only two states – positive versus negative, labeled as 1 and 0, respectively. In the case of a deterministic model, I would obtain a binary result for each message. Theoretically, I could calculate the number of positive and negative messages per minute and assign the dominant state to each minute. Sequentially, I can use such aggregation for each event based on minutes or all message states. However, such an approach is limited and inaccurate when dealing with multiple emotions simultaneously because it ignores non-dominant emotions entirely.

Moreover, this approach fails to account for the spam effect within messages. For example, a message like "Happy Happy Happy" should have a greater impact on the sentiment than a message containing just "Sad". Additionally, there is another problem: a message such as "Happy Sad Happy" would completely ignore the negative sentiment, which is incorrect.

To overcome these limitations, I believe that vector representation allows me to evaluate the versatile sentiment of each message. This representation also helps me adjust the final sentiment coefficient to account for spam and messages with a comprehensive emotional state. Additionally, it helps me to evaluate irony of the message state. Lastly, as a final result, I can provide the separate changes of positive and negative sentiment over time, offering a more nuanced and accurate analysis.

For the base of emotes I would like to detect I used plutchik's emotion model. I believe that in case of chat message emotes should be directly **bipolar**. The original model includes: **Anger vs. Fear, Joy vs. Sadness, Trust vs. Disgust, Surprise vs. Anticipation.**

Emotion	Category	Antonym	Category
Ambiguous	-	Hate	Love
Anger	Calmness	Hope	Despair
Calmness	-	Joy	Sadness
Despair	Hope	Like	Disgust
Disgust	Like	Love	Hate
Anticipation	Surprise	Sadness	Joy
Fear	Calmness	Surprise	Anticipation

Table 1. Emotional categories in SentiSense and antonym

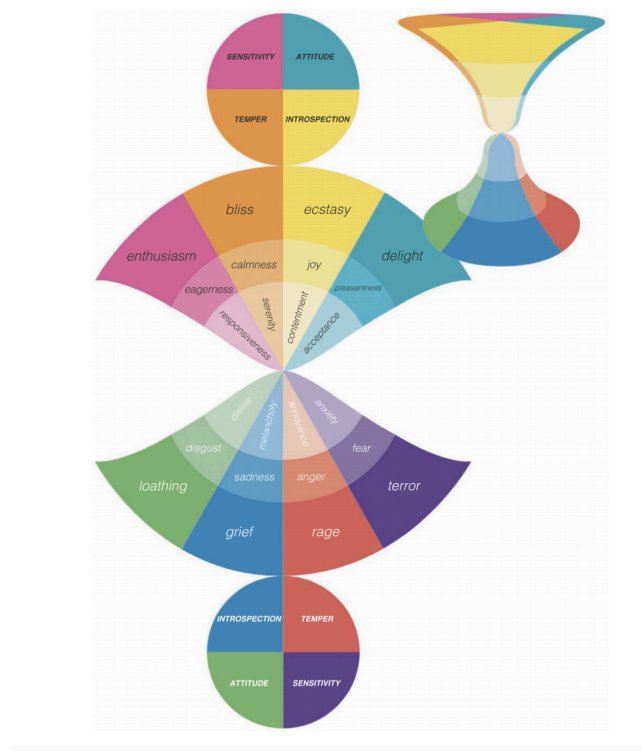


Chart 1. The Hourglass of Emotions

For this project, I would like to use my set of emotions because they are the most explicit, natural and can be well explained and interpreted regarding the vector coefficient in Twitch chats.

<i>Negative spectrum</i>	<i>Positive spectrum</i>	<i>Negative spectrum</i>	<i>Positive spectrum</i>
Anger	Calmness	Anticipation	Surprise
Despair	Hope	Hate	Love
Disgust	Like	Sadness	Joy

Table 2. Set of emotional categories

Due to the nature of Twitch chat (short, brief messages full of emotes and spam), I decided to evaluate sentiment through the emoticons.

There are thousands of emotions at Twitch, more over every streamer has its own emotions. That is why I would like to detect emotions of Twitch chat based on “global Twitch emotes”. For this purpose, I will gather the vector representation of the domain knowledge holders about each emote sentiment representation. Additionally, I would like to compare the “professional” sentiment evaluation with AI models evaluation like Chat GPT and Deepseek and compare the results.

Furthermore, I would like to add two more criteria for respondents:

1. **Confidence** (from 0 to 10). During the analysis of respondents' vectors, I would like to give each respondent a chance to evaluate their confidence in determining each emote sentiment. I can use this parameter during further steps during the creation of the final sentiment vector.
2. **Objectivity** (from 0 to 10). Some emotes are very abstract (for example Doritos chips), that is why I would like to provide an additional coefficient to exclude extra sentiment noises during further steps.
3. **Sarcasm lvl** (from 0 to 10). Sarcasm factor is well-known challenge in sentiment analysis. In case of Twitch chat there are some emotes, that were created as fully sarcastic. That is why I also added the sarcasm factor.

All emotional factors will be evaluated in scale from -10 to +10.

Confidence, objectivity, sarcasm level will be evaluated in scale from 0 to 10.

Confidence and objectivity parameters will approximate the respondents' result by multiplying this coefficient. While sarcasm level would inverse the sign of the vector if the parameter is greater than 5.

The number of words in dataset is 10 242 151. The unique words - 221 113. This shows the high repetition level of words, and the coefficient spam is about 46.32. That means that each word repeats approximately 46 times.

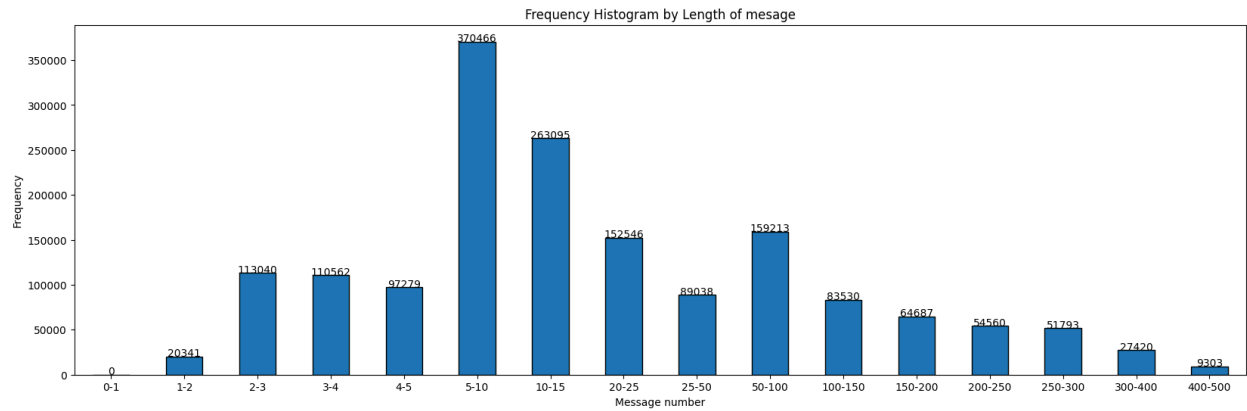


Chart 2. The length of message.

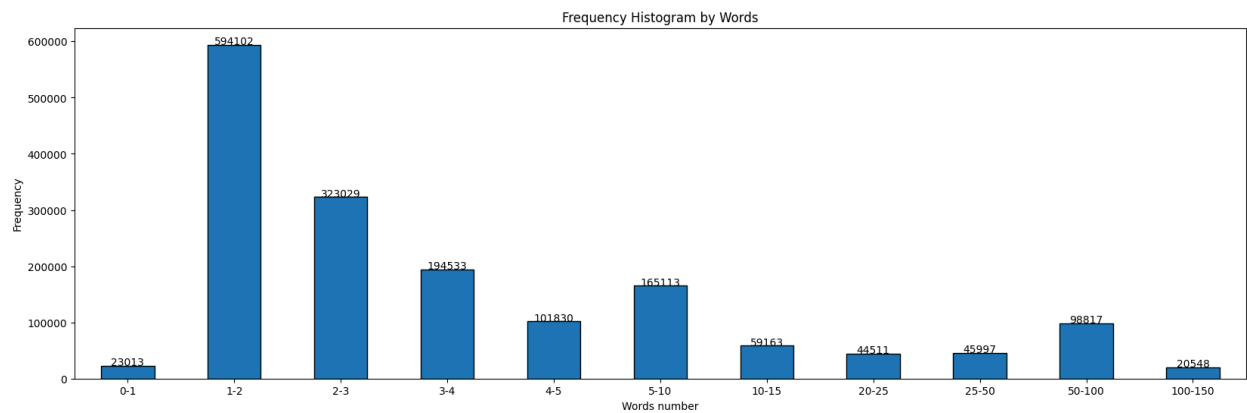


Chart 3. Number of words in message

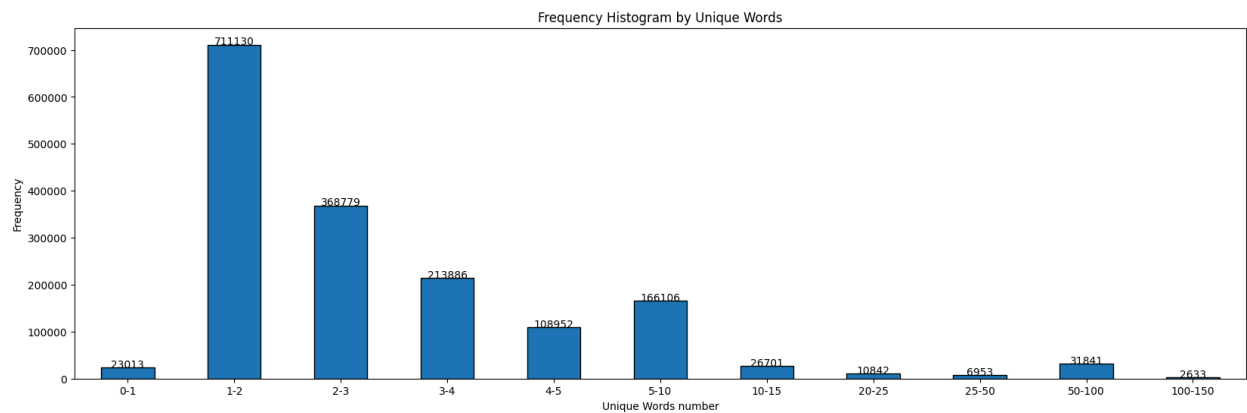


Chart 4. Number of unique words in message

The charts above show that the majority of messages are without context and contain only 1-2 words. Global emots are 12.515% of total words, while global and local emotes 12,526%. The number of messages with at least one global emote is 27.8%. This mean that we can

approximate the more than $\frac{1}{4}$ of chat sentiment based on emoticons only. The number of global emoticons is 314.

First of all, after expert evaluation, I made the average for each feature to make it more objective. I would like to compare human emote descriptions to AI-model emote descriptions. For this, I would use Principal Component Analysis, and plot by the top 2 principal components each emote on the plot. After this, I would use the k-means clustering technique to unite emotes to clusters.

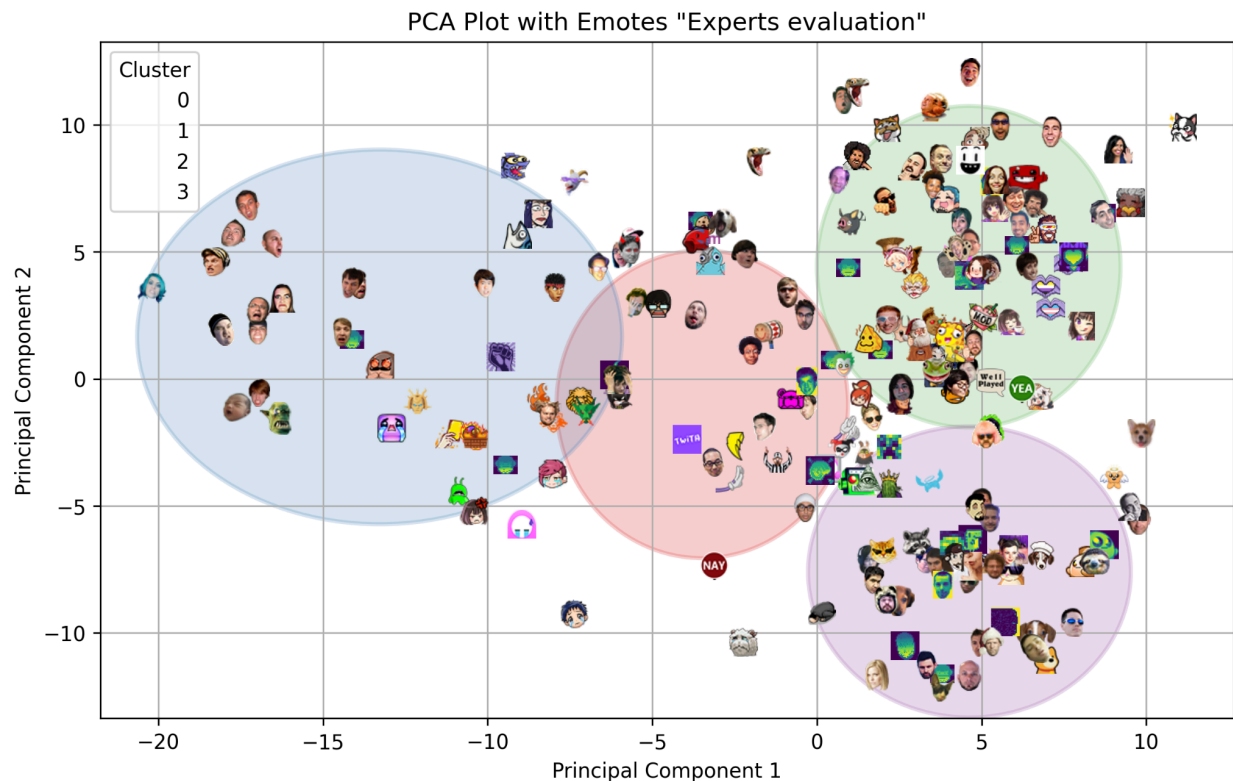


Chart 5. Clustering of emoticons based on expert evaluation

As shown by the analysis, expert annotation revealed more unique results compared to the AI models. The clusters can be described as follows:

Cluster 1 (green): positive, cheerful emotions

Cluster 2 (red): mild surprise, confusion

Cluster 3 (purple): calmness, lack of emotion

Cluster 4 (blue): aggressive, negative emotions

The AI models produced more one-dimensional results, as they do not effectively distinguish between different types of emotions. Based on this, I will proceed by using only the vector-based evaluations provided by human experts.

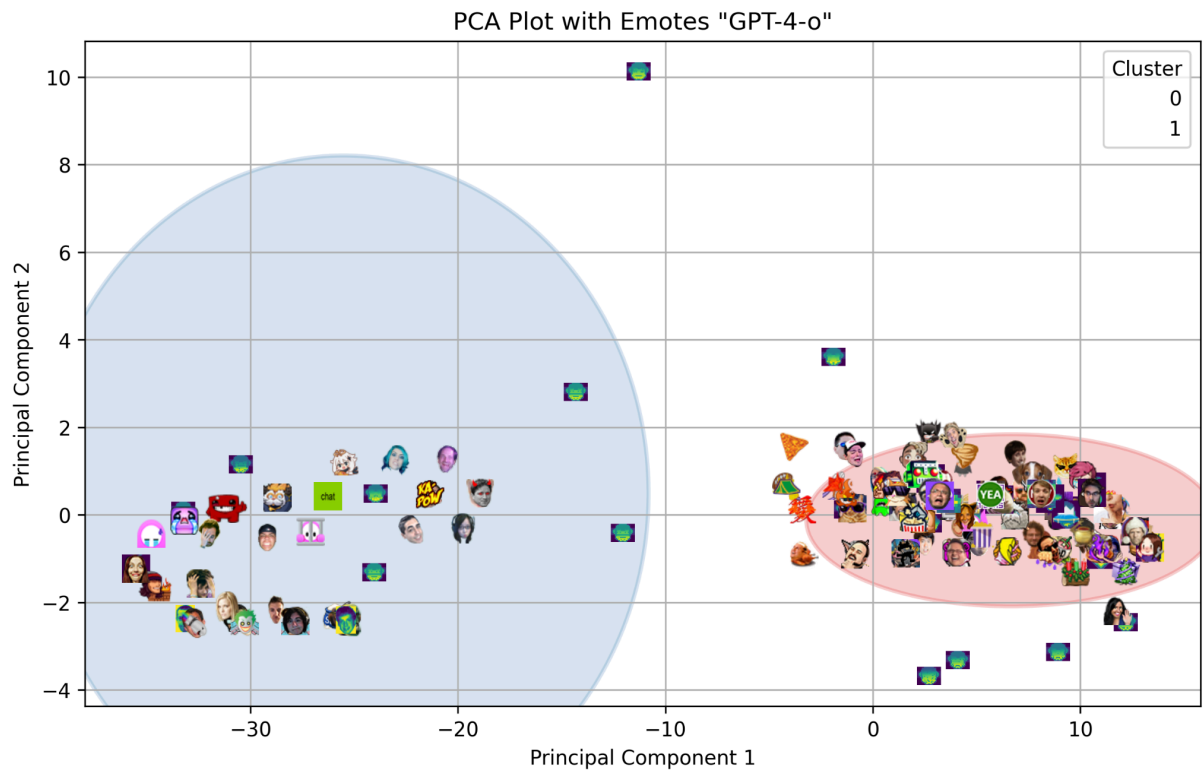


Chart 6. Clustering of emoticons based on GPT-4-o model

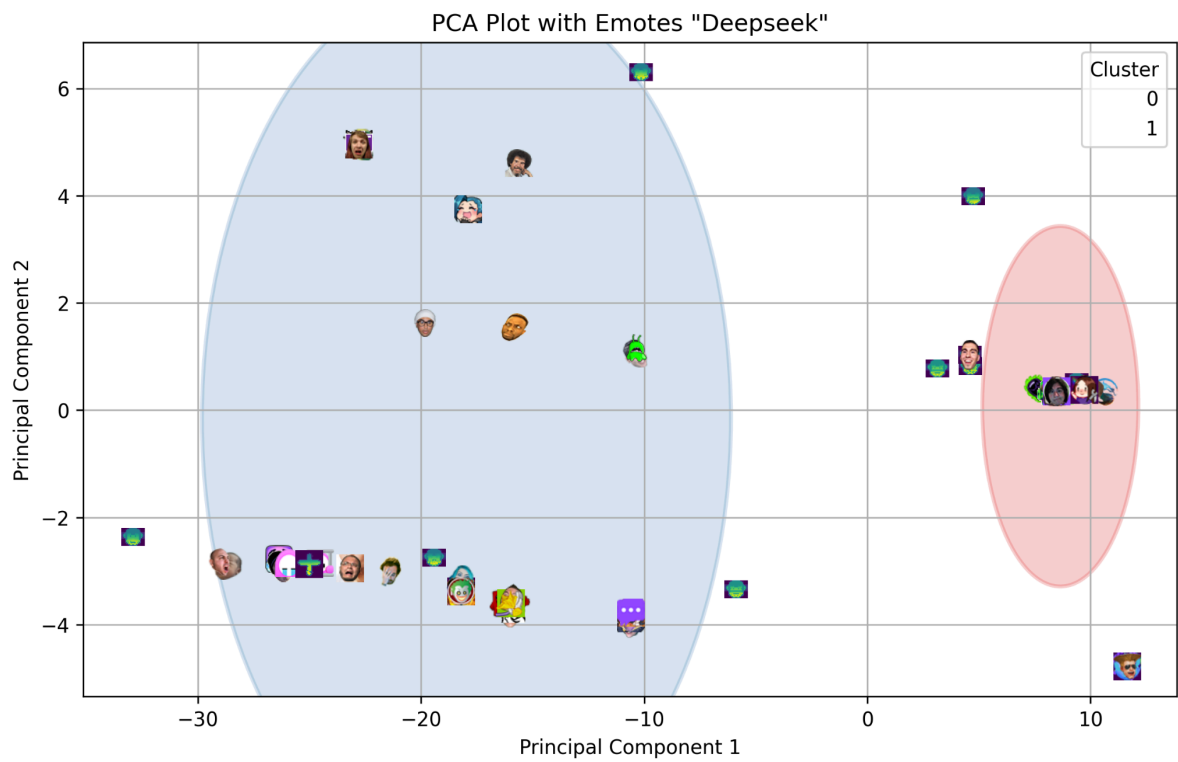


Chart 7. Clustering of emoticons based on Deepseek model

Next steps - creating two dataframes: in each of them, emoticons values will be approximate on coefficient of confidence and objectivity. However, in the first one, the sarcasm feature will be a standalone one. In the second one, sarcasm will inverse the values, as stated above. After the applying this vector representation I would like to compare and find the best approach. As we can see, the overall picture has changed much with the use of the sarcasm coefficient.

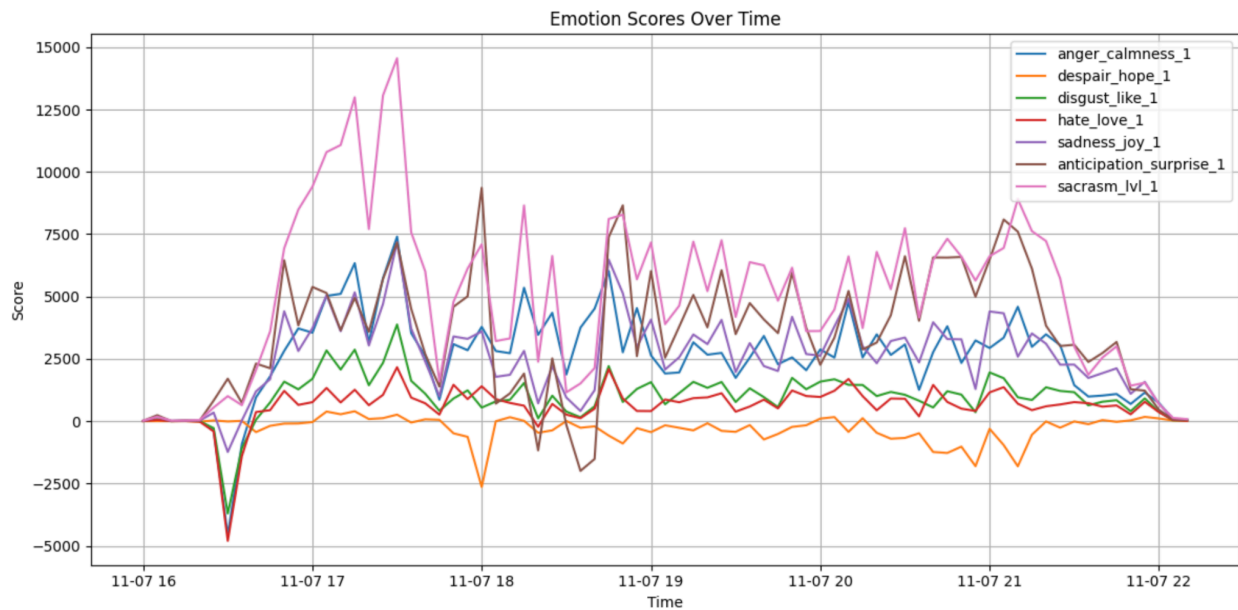


Chart 8. Final day sentiment without sarcasm impact

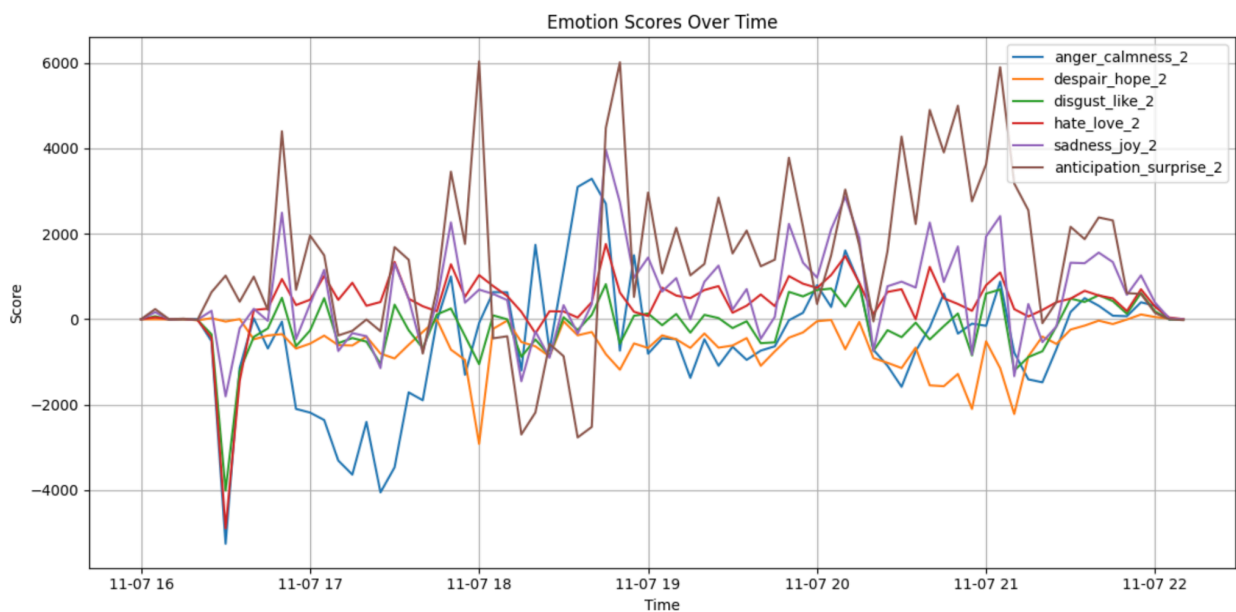


Chart 9. Final day sentiment with sarcasm impact

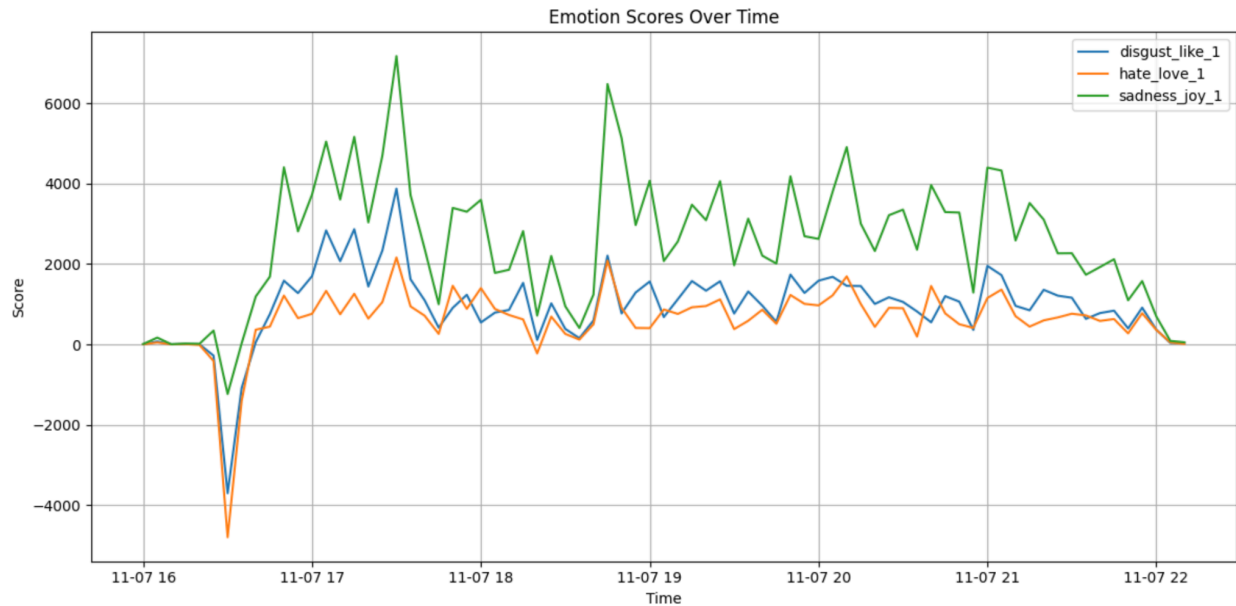


Chart 8.1 Correlated sentiment

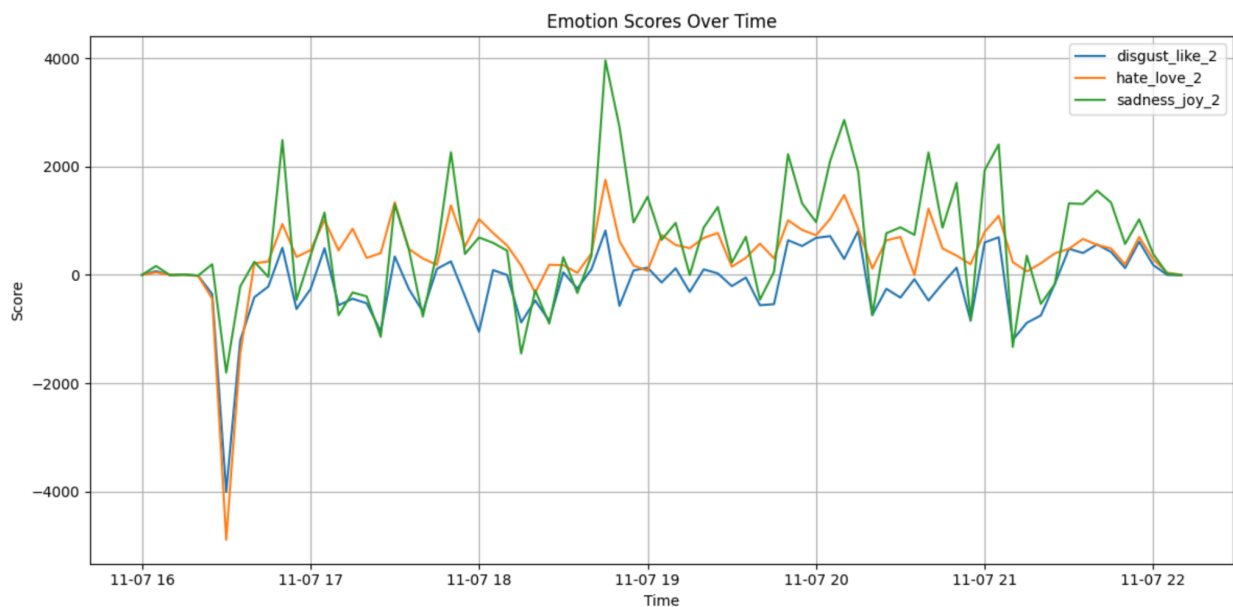


Chart 9.1 Correlated sentiment

When looking at the positive emotion axis **without the sarcasm coefficient**, the graph is mostly positive and rarely dips into negative territory. This could be explained by the fact that viewers tend to express negative emotions less frequently. However, when sarcasm is factored in, the graph shows **much more fluctuation** and often shifts into the **negative range**. Especially in case of disgust and like.

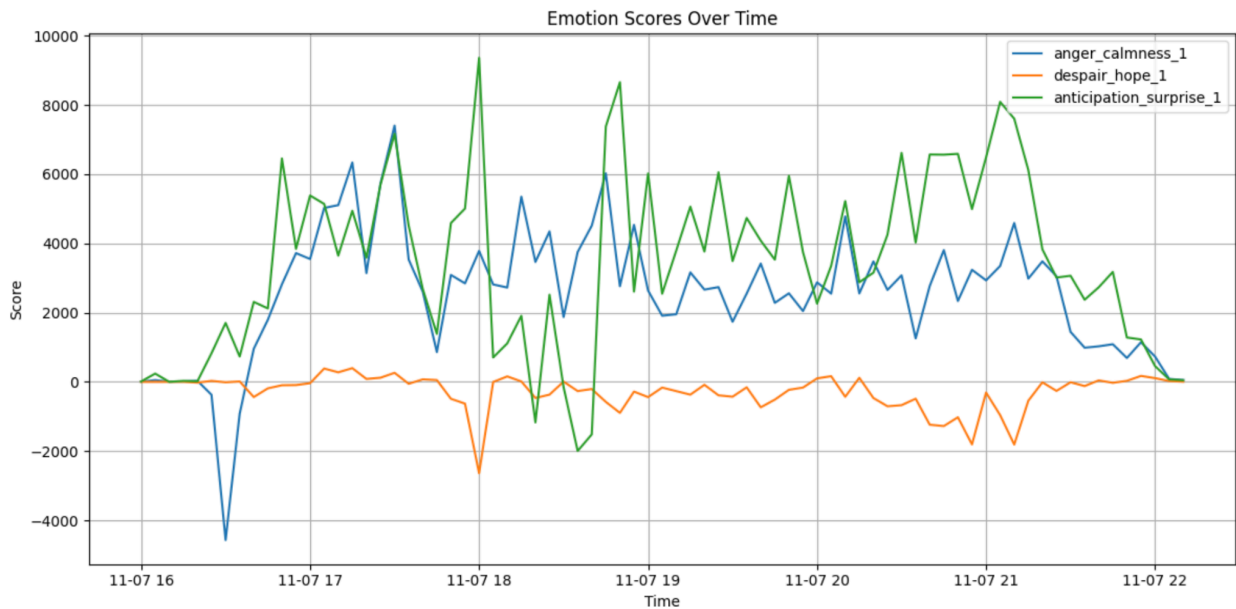


Chart 8.2 Interconnected sentiment

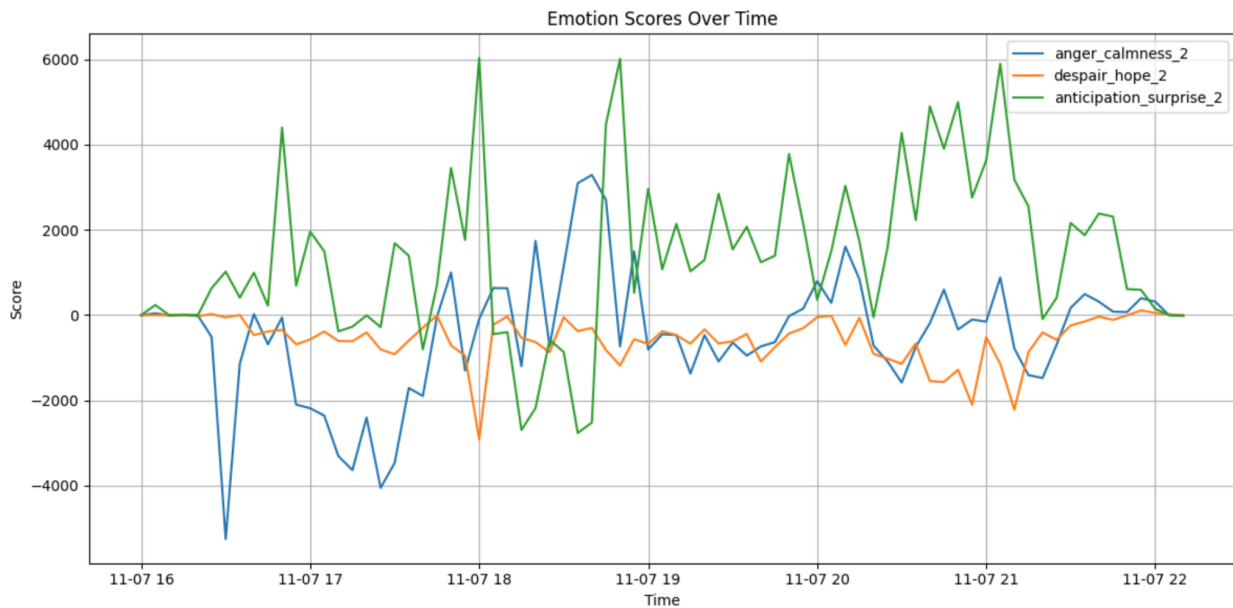


Chart 9.2 Interconnected sentiment

Also, the sarcasm coefficient has very strong impact on the anger and calmness coefficient. The full video with Twitch chat can be found via [Link](#). Now, I am interested in the part around 17:30 GMT 0, to understand what model shows better results. This part was about in one hour after the translation start, and can be found via [Link](#).

The main reason for this downfall was the fact that the sarcasm coefficient reversed the emotive values while this emoticon has slightly calm sentiment, as experts evaluated it, it was the emoticon “OMEGALUL”. This helps me to state, that just reversing all the values with a sarcastic coefficient is not the best idea. Also, as we can see from charts that some of the features are highly correlated.

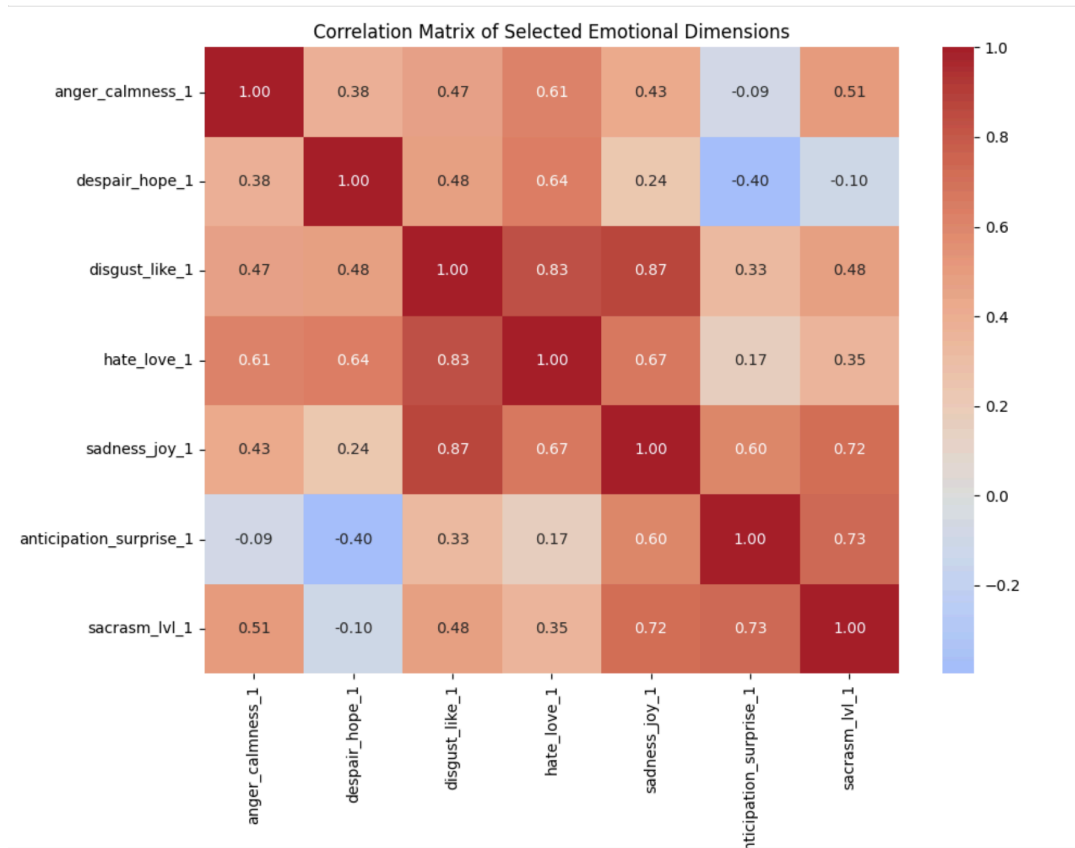


Chart 10.1 Correlation heat map (with sarcasm impact)

For example, the Disgust - Like, Hate - Love, and Sadness - Joy are very correlated between each other. It can be explained that these emotions are very near to each other. Additionally, Anticipation-Surprise, Sadness - Joy are highly correlated with Sarcasm.

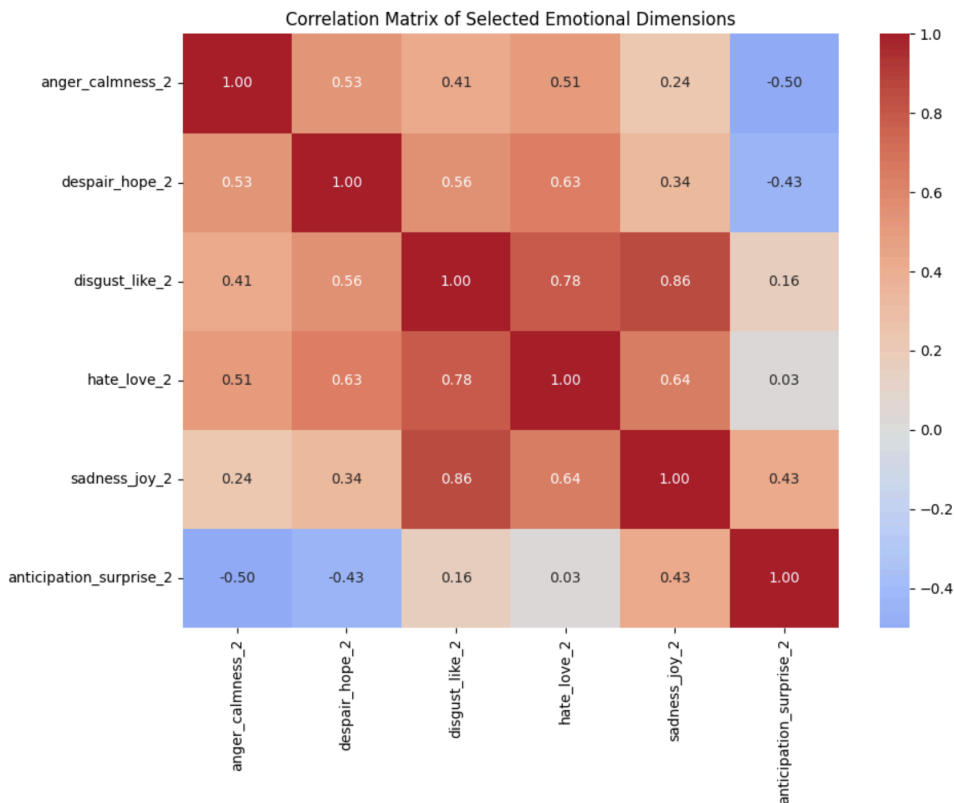


Chart 10.1 Correlation heat map (without sarcasm impact)

Sentiment based on annotators

Audirecognition was provided by elevenlabs.io free service, that can be found via [Link](#). The problems with this service that there was no time stamps at all, that is why I decided primarily to check my hypothesis about how biased annotators' speech. I used for this very simple model TextBlob, to evaluate text as good or bad. There was 1873 good text fragments out of 2173.

Out of samples of bad fragments

- the majority just describe the situation:

"He's doing everything. He truly is the eternal MVP of this game. (rock music plays) Volacy needs something here to get Rifle back. Damone shows up, absolutely stunning play, shutting down Shush and Cadia. (rock music plays"

- some of them can be interpreted as bad, but from perspective of losing team

"All I'm gonna say is this, it's only like 50K less than I've heard a slightly used Bobsky costs."

Sentiment based on main metrics fluctuation

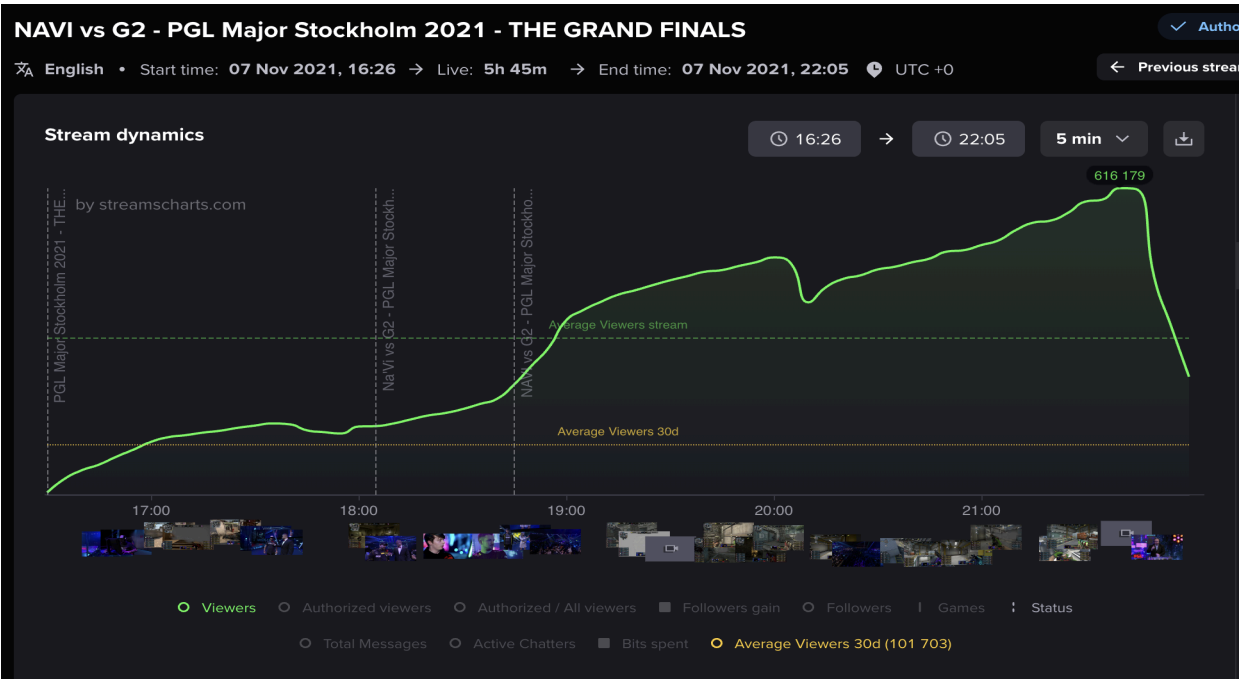


Chart 11. Viewers dynamic

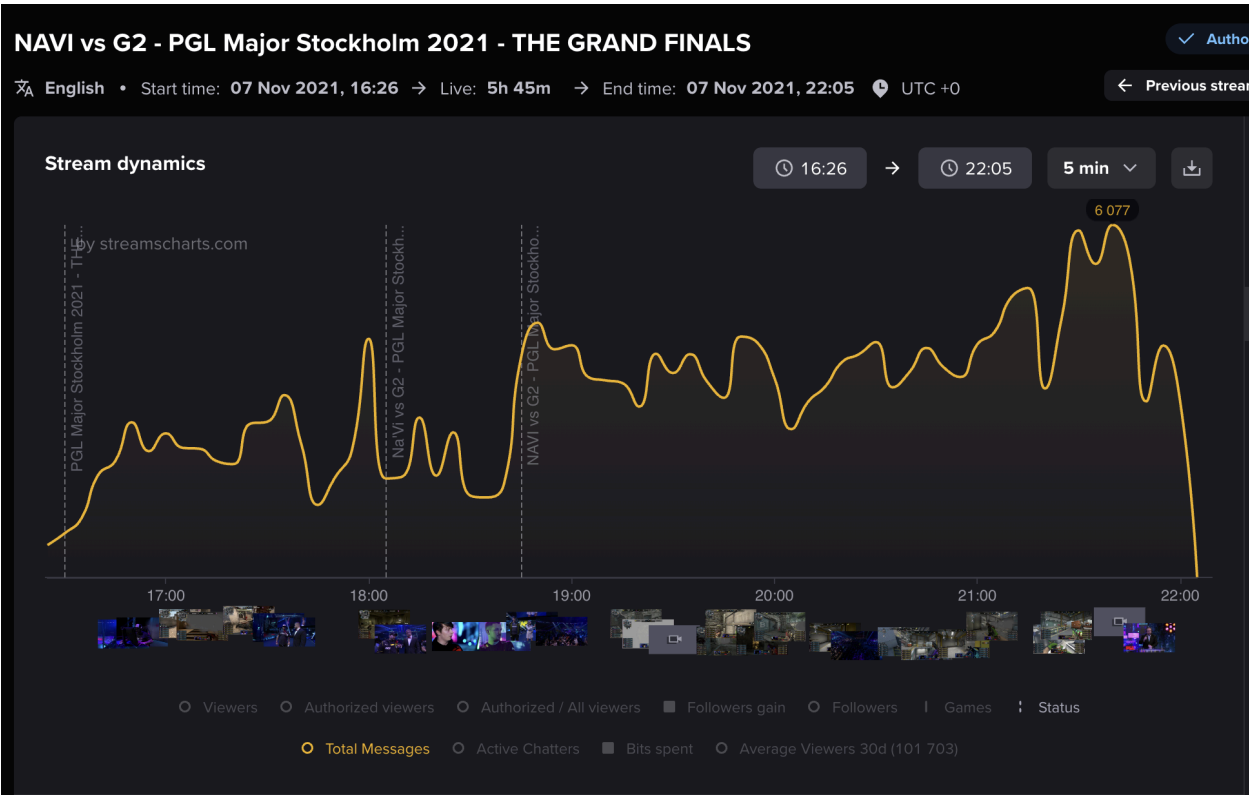


Chart 11. Chat dynamic

I have took the data directly from website, that can be find via the [Link](#).

Audience dynamics clearly show that viewer interest spiked right when the grand final of the tournament began. We also know that the final ended with a 2-0 score (best-of-three format). On the chart, the end of the first match and the beginning of the second match are clearly visible.

Meanwhile, the pre-final All-Star showmatch did not manage to attract a significant audience.

When it comes to chat activity dynamics, the pattern is not as straightforward. It's evident that viewer engagement increased towards the end of the final and towards the end of the All-Star match. However, there are also many local peaks, which likely reflect either highlight moments during the broadcast or moments of direct interaction with the audience.

Concluding remarks

This project explored sentiment analysis in the context of esports livestreams, specifically focusing on Twitch chat data from the "PGL Major Stockholm 2021" — the most viewed Counter-Strike tournament to date. The analysis aimed to capture and interpret real-time viewer sentiment using various approaches: chat messages, annotator commentary, and viewer activity metrics.

A key insight was the uniqueness of Twitch chat as a data source — fast-paced, emote-heavy, and shaped by a highly engaged subset of the audience. To accurately assess the sentiment within this environment, a vector-based approach grounded in Plutchik's emotion model was implemented, augmented with expert evaluations, confidence, objectivity, and sarcasm coefficients. The comparison between expert and AI-generated emotion vectors (e.g., from GPT-4-o and DeepSeek) highlighted that human annotation captured more diverse and precise emotional meanings, especially in the context of sarcastic or ambiguous emotes. Clustering analyses showed how different sentiment clusters emerged depending on the source of annotation and whether sarcasm was factored in.

When sarcasm was incorporated into the sentiment evaluation, the emotional landscape of the broadcast changed significantly — resulting in stronger fluctuations and a noticeable shift toward negative sentiment. Particularly affected were emotions like anger, disgust, and sadness, revealing that sarcasm plays a crucial role in how sentiment is communicated and perceived in Twitch chats.

Furthermore, sentiment inferred from annotator speech showed that most commentary skewed positive or neutral, though occasional fragments reflected the negative sentiment of losing moments or subtle bias. Analysis of viewer activity metrics (such as message volume

and viewership changes) confirmed spikes in engagement during critical moments — particularly at the start and end of the grand final. However, activity patterns were more complex in chat, with numerous local peaks that likely correspond to highlight moments or direct audience interaction.

One way or another, sentiment analysis based on emotes produced promising and fairly logical results. One interesting finding was that some emotional factors are strongly correlated and semantically close, such as like, love, and joy. Another valuable insight was that the emotion of surprise contrasts with calmness and despair, which makes intuitive sense and supports the validity of expert emote evaluations.

Overall, audience sentiment fluctuated throughout the match: at the start of the broadcast, viewers were frustrated and impatient for the game to begin; during the middle, many appeared bored; and toward the end, reactions became more positive.

A particularly noteworthy observation was the high level of sarcasm — viewers frequently used sarcastic emotes, highlighting how deeply irony is embedded in Twitch chat culture.

The last but not least insight is that the viewers use emoticons that reflect their real emotions in super anonymous space as in Twitch chat.

Further steps

The emote-based sentiment evaluation performed surprisingly well, but it currently covers only about 25% of all messages. To improve this approach, the following steps are recommended:

- Involve more expert annotators to make the sentiment evaluation more stable and reliable.
- Supplement the analysis with textual content, not just emotes, to expand coverage.
- Test and implement pre-trained sentiment analysis models (such as those used for tweets or comments) and evaluate how well they perform on Twitch-style textual content.
- Some old emotes are no longer used and not stated as “global”, so for historical data we need to make extra emote analysis.
- Find the best way to work with spammed message, how to account for repetitive emoticons.
- Try different approaches to incorporate sarcastic factor into the model, for example to apply sarcasm on the message level not on the emote level.
- Try to understand the object the viewers talk about.

The sentiment analysis of annotator commentary revealed expected bias, but remains useful mentioning some points:

- It's hard to fully understand the context of what's happening in the game just by looking at viewer chat. To identify key events during peak moments, annotator speech can be essential.

- Add timestamps to commentary data to make it easier to match specific messages with what is happening on screen.
- Text recognition models may struggle to correctly interpret gaming content, especially when it involves slang or jargon. These expressions may be phonetically misrecognized, leading to out-of-context results — and this must be carefully addressed.

The metric-based analysis provides a solid foundation for understanding audience behavior:

- Viewer count trends can be used to identify global peaks in interest.
- Chat spikes highlight local engagement peaks, often aligning with exciting or controversial moments in the stream.

As standalone challenge, I underline multi-language models:

- How annotators and audience react on the same broadcast highlights.
- What is the most discussed topic.
- Is there something that viewers from different countries love and hate the most.

By combining all three approaches — chat sentiment, annotator sentiment, and viewer metrics — it's possible to build a comprehensive, multi-dimensional analysis of the livestream. This would provide clear insight into audience reactions, the context behind them, and statistically supported conclusions.

References

1. Emotion Detection from text: A Survey Lea Canales, Patricio Martínez-Barco Department of Software and Computing Systems University of Alicante